To Be Submitted

GitHub repository	**rheijovenjuan/AI_231** — public, MIT
Dataset location	**markandrian30/AI231** (GitHub, `MEX2/OptionB`) — no DOI; cite repo + access date **2026-09**; real-voice + wake clips committed in this repo
A100 cluster	**n003.ai.internal** · 1 × A100-SXM4-40GB · `n_gpu=1` · ≈15 min/run · seeds **1234** (data 20260929)
Model weights	`output/onnx/` (wake/keyword/command.onnx + model_card.json) and `output/training/checkpoints/` — MIT, committed

What the reviewer will look for

#	Item	Status
1	Repo public, one-command reproduction	**done — ./setup_pi.sh · python app.py · run_keyword.sh**
2	Dataset licensed and citable (DOI)	**done — source repo + access date, no-DOI note (SUBMISSION §1)**
3	Training logs + final checkpoint committed	**done — output/training/{logs,checkpoints,history}/**
4	Pi 4 latency reproduced by the posted script	**done — runtime/benchmark.py --label rpi4 --threads 2 → benchmark_rpi4.json**
5	Held-out test set, unseen speakers	**done — speaker-disjoint 70/15/15; test s91–s100**
6	Baseline of comparable size compared	**done — 123 303 vs 123 496 params · 99.84 % vs 99.29 %**

ME2 — Voice Controlled Smart Device

Always-on keyword + intent recognition at the edge · no cloud round-trip

2 October 2026 · Rhei Juan (rheijovenjuan) · Model · Dataset · Training on A100 · Validation on RPi 4 + PC

Model

Mic · 16 kHz → log-mel 40 × 250
2.5 s window · HTK mel 20–7 600 Hz · CMVN (wake: 40 × 120)

DS-CNN encoder · widths 64→192 · 192-d
MobileNetV2-style depthwise-separable · full-window (non-causal)

Intent head
40 keyword sigmoids → rule → 19 intents

Slot head
rule over keyword set · 6 slots, 19 values + NONE

Actuator
wake ≥ 0.900 → ack beep → app.py action

Raspberry Pi 4
onnxruntime 1.30.0 · 2 threads · ~1.1 MB payload

Item	Value
Parameters / weights	**0.27 M · 1.1 MB**
wake / keyword / command	23 665 / 123 496 / 123 303

Validation on the Raspberry Pi 4

Item	Value
Keyword / intent acc	**99.29 % / 99.46 %** (test, n = 1 842)
False-accept rate	**0.000 %** wake FAR @ 0.900 (0/1 904) · streaming FA 0/19
Latency wake+cmd / RTF	**69.3 ms** / **0.016**

Dataset

Item	Value
Source	markandrian30/AI231 (OptionB) + 94 real-voice clips
Hours / utts	**8.8 h / 18 469**
Speakers	100 TTS + 1 real (test s91–s100)
Labels	19 intents · 6 slots · 40 keywords
Splits (spk-disjoint)	14 752 / 1 875 / 1 842

Training on the A100 cluster

Item	Value
Cluster	1 × A100-SXM4-40GB (n003), n_gpu=1
Objective	BCE (40-kw multi-label) → rule · CE baseline · BCE wake
Optimiser	AdamW + OneCycleLR (max lr 1.6e-3 → 0)
Steps / loss	**60 epochs (best 50) / 0.0345** · seed 1234 · ≈15 min
Run-5 retrain	+94 real clips → test intent 99.24→**99.29 %**

Headline accuracy

metric	value
keyword head — intent (test)	**99.29 %**
keyword head — slot (test)	**99.46 %**
baseline softmax — intent	99.84 %

Runtime	onnxruntime 1.30.0 · 2 thr

real-voice slice (val, was 42.1 %)	78.9 %
wake TPR @ 0.900 (test)	99.54 %

Docs: ACCURACY.md · BENCHMARKS.md · poster_data.md

Benchmarks — Pi 4 vs PC

quantity	Pi 4 (p50/p99)	PC (p50/p99)
model load	649 ms	221.6 ms
wake inference	17.2 / 32.1 ms	3.4 / 6.5 ms
keyword inference	40.0 / 70.7 ms	9.7 / 11.9 ms
wake+command end-to-end	**69.3 ms**	**16.9 ms**
peak RSS	147.0 MB	139.6 MB
real-time factor	0.0160	0.0036

Measured 2026-10-02/03 · runtime/benchmark.py --threads 2 (Pi) · raw: output/onnx/benchmark_{rpi4,pc}.json